

LoCoBench: A Benchmark for Logical-Coherence Scorers

Phase 1 — facets, error taxonomy, generation prompts, gold schema and metrics

FactReasoner — Ideation

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Abstract

The logical-coherence pipeline mines thirteen Level-2 discourse senses between the atoms of a response, compiles them to five Level-1 Markov-network couplings, and reads a Logical Coherence Score (LCS) off the resulting graph. None of it is evaluated on real data: what exists today is nine hand-authored fixtures and one exact-value regression oracle. This document designs the benchmark that would fix that. It (i) takes the construction methodology of LOREFACT [LoReFact] — the first long-form fact-checking dataset with logical annotations — and identifies precisely what its four-relation inventory can and cannot express about ours, finding that *three of our five factor tables have no existing gold data anywhere*; (ii) defines LoCoBench’s *facet* system — the structural axes a datapoint is graded on, kept deliberately disjoint from the 36 subject *topics* it inherits wholesale from LOREFACT and must cover in full — together with target counts and a deliberate departure from LOREFACT’s 78%-false corpus balance, which a graded score cannot use; (iii) defines a ten-operator perturbation taxonomy with, for each operator, a *per-readout* expected direction, including two strong falsifiable invariance tests that the taxonomy’s own structure licenses; (iv) gives the complete generation and validation prompt suite, copy-ready, extending LOREFACT’s Tables 11–14 and 20–25 and replacing its prose “logical statement” stage with a machine-readable relation plan; (v) specifies the gold schema as a strict superset of the shipped fixture format; and (vi) defines the metrics for both gradable targets, with LOREFACT-comparable analogues. Scoring is scoped to three readouts — `mean_marginal`, `consistency` and `log_partition`, one each for belief, event activity and mass — for the reasons given in Section 5.1. Two findings shape the design and are stated up front, because a benchmark that ignored either would silently test the wrong thing: the readouts are not jointly monotone under increasing coherence — one of them provably inverts at the concession rung — and the taxonomy’s only fixed strength prior is unreachable on the production code path.

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1 Motivation, and what LoReFact does and does not give us

1.1 The evaluation gap

The shipped coherence stack (`src/fact_reasoner/lcs/`) has a relation miner, a two-level taxonomy, a Markov-network builder shared with the factuality assessor, and four score readouts. Its empirical basis is:

- nine hand-authored fixtures in `data/lcs/*.json`, carrying atoms and prose `notes` but *no machine-readable gold relations*;
- one regression oracle — the AeroParts model, whose exact values (`mean_marginal` = 0.587, `log Z` = −9.75, `consistency` = 0.813, `log_partition` = 0.7831) are pinned by a test;
- an evaluation *plan* (`research_plan.md` §9) naming perturbation ladders, Spearman-versus-human correlation, and three baselines [[DiscoScore](#), [G-Eval](#), [SelfCheckGPT](#)] — but no dataset, no facet inventory, and no gold format.

So we cannot currently answer the two questions that matter: *does the miner recover the relations a response actually draws?* and *does the LCS order responses by coherence?* This document designs the data that would answer both.

1.2 LoReFact

LOREFACT [[LoReFact](#)] is the closest prior art and the right starting point. It argues that the dominant decompose-then-verify fact-checking paradigm verifies atomic claims individually and overlooks the *logical dependencies* among them, so a text that combines true facts through a false dependency is misjudged as factual. Its preliminary study found that SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of logic-related samples respectively.

Its dataset contributions, which we reuse:

- **A controlled three-stage generation pipeline:** for each topic, generate a logic-eliciting question; generate 20 atomic claims (10 correct, 10 incorrect); compose at least two *logical statements* from them; then draft a response that naturally embeds all of the statements. Full prompts in their Tables 11–14.
- **Function-call-style annotation prompts** (their Tables 20–25) for decomposition, checkworthiness filtering, logic-type detection and verification — a house style we adopt for our validation prompts in Section 6.
- **Scale and coverage:** 1,080 samples, 5,706 annotated logical statements, 36 topics across 4 domains, average response length 819 words, 3–7 logical statements per sample, inter-annotator agreement 92.18%.

1.3 Why extending it is not incremental

LOREFACT is built for a different task, and the mismatch is structural (Table 1).

The consequence, developed in Section 3, is that LOREFACT’s label set reaches 7 of our 13 senses and 2 of our 5 couplings. EQUIVALENCE, EXCLUSIVE and CO-NECESSITY — three of the five pairwise factor tables the network can build — have **zero** gold data in any existing dataset. That is the strongest single argument for building this benchmark rather than annotating on top of theirs.

	LoReFact	LoCoBench (this design)
Relation inventory	4 types: causal, conditional, temporal, concessive (+ <code>no_logical_relationship</code>)	13 Level-2 senses compiling to 5 Level-1 couplings (+ NONE)
Relation label	binary logic-factuality per statement	graded $p = P(\tau a_i, a_j) \cdot P(a_j a_i, \tau)$
Score granularity	per-statement, arithmetic mean	graph-level readout off a Markov network
Direction	not distinguished	directed per sense; Cause-Effect vs. Effect-Cause
Exhaustiveness	not represented	CONTRADICTION vs. EXCLUSIVE turns on it
Resolution	not represented	Concession carries a resolver and a discount
Corpus balance	78% logically false by design (4,429 / 5,706)	must also be correct at the <i>coherent</i> end

Table 1: The gap between LOREFACT’s design and what a logical-coherence scorer needs. Each row is a dimension along which their gold labels cannot grade our system.

2 Two findings that shape the design

Both were verified against the source while designing, and both are stated before the design because a benchmark that assumed otherwise would silently measure the wrong thing.

2.1 Finding 1: the readouts are not jointly monotone — by construction

`research_plan.md` §9 states the natural requirement: “a good LCS must drop monotonically with perturbation severity.” Applied to every readout alike, *this requirement is false*, and a benchmark enforcing it would fail a correct implementation.

The MRF deep-dive (§“The shared concession quirk”) establishes why. The concession edit *softens* a resolved conflict from $p = 0.80$ to $p = 0.55$. A weaker EXCLUSIVE (or CONTRADICTION) factor puts *more* mass on the both-true cell — its (1, 1) entry is $1 - p$ — so for a lone edge under uniform priors

$$P(a_{11}=1, a_{12}=1) = 0.10 \quad (p=0.80) \quad \longrightarrow \quad 0.225 \quad (p=0.55).$$

The readout that asks “*is the conflict active*” — **consistency**, which counts the event — therefore sees the softened edge as *more* often active and **dips**. Readouts that ask “*how strongly do we believe each atom*” — **mean marginal** on the marginals, **log partition** on the mass — correctly read the softening as *less* suppression and **rise**.

Two design consequences, carried through Sections 4, 5 and 8:

- (1) Ordering contracts are **per readout**, not global (the C1/C2/C3 classes of Section 5.3).
- (2) The inversion becomes a **positive prediction** rather than an exemption: a system whose **consistency** is monotone across the concession rung is *not* implementing the concession discount. Section 8.3 makes this the *quirk-conformity* metric, whose target is high.

2.2 Finding 2: Restatement’s strength prior is unreachable in production

`COMPILE[RESTATEMENT].strength_prior = 0.90` is the taxonomy’s only fixed prior, and `compile_sense` does use it — but only when its `raw_p` argument is `None`. The single call site in the miner is

```
compiled_level1, _, spec = compile_sense(sense, None)
```

which passes `None` and discards the effective-strength return value entirely (the `_`). Strength on the production path comes from Prompt B alone. The 0.90 prior therefore never reaches a factor table.

So the benchmark *cannot* test the prior through the pipeline. What it can test is whether the prior is *empirically justified*: whether mined strength on gold Restatement edges concentrates near 0.90. Section 9 records this as an open item (R4) rather than a metric, so no one later writes an assertion that silently passes.

3 The LoCoBench facet system

3.1 The facets of LoCoBench

Terminology, and a collision to avoid. LOREFACT organizes its corpus by *subject matter*: 36 **topics** (Literature, Astronomy, Civil Engineering, ...) grouped into 4 **domains** (Table 6). LoCoBench inherits that grid wholesale — every one of the 36 topics must be represented (Section 3.2) — so those two words are **reserved throughout this document for subject matter, and never used for anything else.**

What LoCoBench adds is a second, orthogonal organization: the structural properties a data-point is labelled with and graded on. We call these **facets**. A *facet* is one controlled vocabulary; a *facet value* is a value drawn from it. The two organizations are independent by construction — a single item has a topic *and* a full set of facet values, and the benchmark’s coverage requirements apply to each separately.

	Subject organization (inherited)	Facet system (LoCoBench’s own)
Terms	<i>topic, domain</i>	<i>facet, facet value</i>
Cardinality	36 topics in 4 domains	3 facet groups, 19 facets
Answers	“what is this text <i>about</i> ?”	“what is being <i>tested</i> here?”
Source	LOREFACT Table 9, adopted unchanged	new; derived from <code>lcs/taxonomy.py</code> and this design
Requirement	all 36 topics represented (§3.2)	per-facet targets (Tables 3–5)

Table 2: The two orthogonal organizations of the corpus. Keeping the vocabularies disjoint matters because both are coverage requirements and both are reported: a result broken down by *topic* says whether the benchmark generalizes across subject matter, while a result broken down by *facet* says which structural phenomena a scorer handles. Conflating them would make neither breakdown interpretable.

There are three groups of facets, listed here in one place so the rest of the document can refer to them without re-enumerating:

- (1) **Relation facets** — the Level-2 *senses* and the Level-1 *couplings* they compile to (Table 3). These are what a gold edge is labelled with, and what the miner is graded on.

- (2) **Item facets** — the axes along which items and families vary: family type, rung, perturbation operator, ordering contract, split (Table 4).
- (3) **Annotation facets** — the enumerated values the schema’s own fields take (Table 5). Each is defined operationally in a prompt or a schema note; that table is the index to them.

Within the relation facets, *sense* and *coupling* always name the two levels specifically, and are never used interchangeably with each other or with “facet”.

(1) **Relation facets.** Thirteen senses compiling to five couplings plus NONE. The **coverage** column is the distinction that motivates the benchmark: which of these relations already have gold data somewhere, and which do not.

Level-2 sense	Level-1 coupling	Coverage	Role in the benchmark
<i>Positive couplings — the relation supports its target</i>			
Cause-Effect	ENTAILMENT	covered	the modal relation of explanatory prose
Effect-Cause	ENTAILMENT	covered	direction partner of Cause-Effect
Evidence	ENTAILMENT	partial	collapsed into “causal” by LOREFACT
Condition	ENTAILMENT	covered	confusable with Precedence
Instantiation	ENTAILMENT	new	general↔specific; no prior gold
Restatement	EQUIVALENCE	new coupling	the only sense with a fixed prior (0.90)
<i>Conflict couplings — define the “active conflict” event</i>			
Contrast	CONTRADICTION	new	non-exhaustive opposition
Concession	CONTRADICTION	new	a tension the text resolves; 0.45 discount
Alternative	EXCLUSIVE	new coupling	<i>exhaustive</i> opposition: exactly one holds
<i>Disjunctive coupling — not a conflict</i>			
Disjunction	CO_NECESSITY	new coupling	at least one holds; penalizes only (0,0)
<i>No coupling — no factor is added to the network</i>			
Precedence	NONE	none	ordering only; invariance target
Succession	NONE	none	ordering only; invariance target
None	NONE	none	the distractor label; over-connection denominator

Table 3: **The 13 relation facets of LoCoBench**, grouped by what their coupling does to the Markov network. **new** no prior gold data anywhere **partial** covered, but only coarsely **none** no MRF factor (NONE). Six senses are new: Instantiation, Restatement, Contrast, Concession, Alternative, Disjunction — and three of the five couplings (EQUIVALENCE, EXCLUSIVE, CO_NECESSITY) have *no gold data in any existing dataset*, which is the case for building this benchmark (Section 3.3). Property flags (directed, concession, ordering-only) and the normative compilation map are in Table 8; per-sense target counts in Table 9.

(2) **Item facets.** The axes along which items and families vary. Family type and rung together determine what a given item is *for*; the operator says how it was derived from its parent rung.

(3) **Annotation facets.** These are the enumerated values the schema’s fields take (Section 7). Each is defined where it is operationally used — in a prompt’s instructions, or a schema field note — so this table is the index rather than the definition.

Axis	#	Values
Family type	5	F-CONF, F-CHAIN, F-EXH, F-ORDER, F-INV (Table 14)
Rung	5	worse, base, concession_resolved, fix_one_conflict, coherent (Table 13)
Perturbation op.	10	P-WS, P-DR, P-EF, P-CR, P-IC, P-BC, P-OS, P-SR, P-DL, P-OO (Table 11)
Ordering contract	3	C1 full chain, C2 quirk chain, C3 endpoints (Section 5.3)
Confusable pair	11	weighted 1–3 (Table 10)
Readout	3	mean_marginal, consistency, log_partition (Section 5.1)
Split	2	dev (the 9 shipped fixtures), test (generated) (Section 5.5)

Table 4: **Item facets**: the axes along which benchmark items and families vary. Each row cites the section that defines it normatively. Subject matter is deliberately *not* listed here — topic and domain are the inherited organization of Section 3.2, orthogonal to every facet above (Table 2).

Field	#	Values	Defined in
validity	2	valid, invalid	P3, instr. 8
error_kind	4	wrong_sense, wrong_direction, false_endpoint, spurious	P3, instr. 8
strength_band	3	strong [0.85, 1.00], moderate [0.60, 0.84], weak [0.35, 0.59]	P3, instr. 7
role (atom)	3	claim, holding, distractor	§7
window_admission	4	window, gate, discourse_promoted, out_of_window	§7
readout_directions	4	decrease, increase, invariant, unconstrained	§7, Table 11
Claim tag (P2)	9	[correct], [incorrect], [alt-pair-1/2], [disj-pair-1/2], [equiv-pair-1/2], [holding]	P2, instr. 7
V2 verdict	4	A exhaustive, B non-exhaustive, C disjunction, D neither	V2
V4 status	4	asserted, altered, missing, merged	V4

Table 5: **Annotation facets.** Two of these are load-bearing outside their defining prompt and are easy to miss: `error_kind` is a required gold field, and the `strength_band` ranges are what metric A7 (Section 8.1) grades against. Prompt references are to Appendix A.

3.2 Subject coverage: all 36 LoReFact topics

Requirement. LoCoBench must contain datapoints from **every one of the 36 topics** of LOREFACT Table 9, reproduced here as Table 6. This is a hard coverage constraint, not a sampling preference: it is what makes a LoCoBench result comparable to a LOREFACT result on the same subject matter, and what prevents the facet targets of Section 3.1 from being met inside a narrow slice of subject matter that happens to suit them.

Allocation. With 120 families over 36 topics, the target is $\lfloor 120/36 \rfloor = 3$ families per topic plus 12 families distributed to the topics that best support the scarce relation facets. Section 6 biases that surplus toward adjudicated subject matter — incident investigation, litigation, diagnostic reasoning — which is where exhaustive alternatives and resolving holdings occur naturally. Concretely the surplus goes to Law, Medicine, Civil Engineering, Political Science and Ethics, while the **3-family floor holds for all 36 topics without exception.**

Reporting. Every headline metric of Section 8 is reported both pooled and broken down by domain, as LOREFACT does (their Table 3). A per-*topic* breakdown at 3 families per topic is too thin for a stable estimate, so topic coverage is verified as a *constraint* on the corpus rather than reported as 36 separate scores.

3.3 Mapping LoReFact’s inventory onto ours

Table 7 maps each LOREFACT type onto the senses it covers.

Natural Sci. (11)	Humanities (10)	Engineering & Tech. (8)	Social Sci. (7)
Astronomy	Archaeology	Artificial Intelligence	Anthropology
Biology	Art	Civil Engineering	Economics
Botany	Culture	Computer Science	Education
Chemistry	Ethics	Cybersecurity	Law
Geology	Gender Studies	Electrical Engineering	Political Science
Human Biology	History	Mechanical Engineering	Psychology
Medicine	Linguistics	Renewable Energy Tech.	Sociology
Meteorology	Literature	Robotics	
Oceanography	Philosophy		
Physics	Religion		
Zoology			

Table 6: **The 36 subject topics**, adopted unchanged from LOREFACT Table 9 and grouped by their 4 domains. Every topic must contribute at least one family. Note that LOREFACT’s own example tables (their Tables 16–19) name five of these differently — *Animals*, *Human Body*, *Ocean*, *Plants*, *Weather* for *Zoology*, *Human Biology*, *Oceanography*, *Botany*, *Meteorology* — and two topics appear there with a duplicated example question. We treat Table 9 as authoritative and record the variance so a Phase-2 topic list can be diffed against either.

LoReFact type	Covers our senses	n	Mismatch
causal	Cause-Effect, Effect-Cause, Evidence	3,358	direction not distinguished; Evidence (epistemic) collapsed into causal (ontic)
conditional	Condition	1,551	1:1
temporal	Precedence, Succession	473	semantic mismatch: treated as truth-bearing, whereas we compile both to NONE (ordering_only)
concessive	Concession	324	no resolved/unresolved distinction; conflated with Contrast
no_logical_rel.	None	—	1:1
<i>Senses LOReFACT structurally cannot express — no gold data exists for these</i>			
(none)	Alternative $\rightarrow$ EXCLUSIVE	0	no type encodes “exactly one holds” — the coupling penalizing <i>both</i> same-value worlds has no analogue in any prior dataset
(none)	Disjunction $\rightarrow$ CO-NECESSITY	0	their label set has no positive-polarity multi-claim coupling at all
(none)	Restatement $\rightarrow$ EQUIVALENCE	0	their claim prompt requires claims be “basic, indivisible, and independently verifiable”, which <i>forbids</i> the paraphrase pairs an equivalence edge needs
(none)	Instantiation	0	general $\leftrightarrow$ specific, silently absorbed into causal
(none)	Contrast (vs. Concession)	0	their single “concessive” type conflates unresolved non-exhaustive opposition with a resolved tension
(none)	Concession <i>with resolution</i>	0	the is_concession + resolver structure driving the $p_{\text{eff}} = p(1 - 0.45)$ discount is unrepresented, so the discount is untestable against their data

Table 7: LOReFACT $\rightarrow$ LoCoBench sense coverage, with their statement counts (their Table 4). **Upper block:** the seven of our thirteen senses their four types reach, and how coarsely. **Lower block, shaded:** the six they cannot express, with a statement count of zero — these are why EQUIVALENCE, EXCLUSIVE and CO-NECESSITY have no gold data anywhere. new no prior gold data anywhere partial covered, but only coarsely none no MRF factor (NONE).

3.4 The normative compilation matrix

Table 8 reproduces COMPILE from `src/fact_reasoner/lcs/taxonomy.py`. **This table is normative:** a benchmark item whose `level1_coupling` disagrees with `COMPILE[level2_sense].level1` is malformed, and Section 7 requires a builder assertion to that effect.

Level-2 sense	Level-1 coupling	prior	directed	concession	ordering-only
Cause-Effect	ENTAILMENT	—	yes	—	—
Effect-Cause	ENTAILMENT	—	yes	—	—
Evidence	ENTAILMENT	—	yes	—	—
Condition	ENTAILMENT	—	yes	—	—
Instantiation	ENTAILMENT	—	yes	—	—
Restatement	EQUIVALENCE	0.90	no	—	—
Contrast	CONTRADICTION	—	no	—	—
Concession	CONTRADICTION	—	yes	yes	—
Alternative	EXCLUSIVE	—	no	—	—
Disjunction	CO_NECESSITY	—	no	—	—
Precedence	NONE	—	yes	—	yes
Succession	NONE	—	yes	—	yes
None	NONE	—	no	—	—

Table 8: The Level-2 $\rightarrow$ Level-1 compilation map (deep-dive Table 2), verbatim from `taxonomy.py`. Shading repeats the coverage classes of Table 3 (`new` no prior gold data anywhere `partial` covered, but only coarsely `none` no MRF factor (NONE)) so the property matrix and the inventory can be read together. Restatement is the only sense with a fixed strength prior — and see Finding 2 (Section 2.2) for why that prior is not reachable in production. Conflict couplings, which define the “active conflict” event for **consistency**, are exactly {CONTRADICTION, EXCLUSIVE}; CO_NECESSITY is a *positive* coupling and is not a conflict.

Two boundaries in this table carry the design’s difficulty, and both need explicit annotation to be gradeable at all:

Contrast vs. Alternative — non-exhaustive versus exhaustive opposition. Contrast forbids only the both-true world; Alternative forbids both-true *and* both-false. This is the sole distinction in the taxonomy that moves a relation between CONTRADICTION and EXCLUSIVE, i.e. between two different factor tables. Deciding it requires quantifying over possible worlds (“could both be false?”), which is why Section 6 gives it a dedicated adjudication prompt with an explicit decision procedure, and Section 9 rates it our highest risk.

Contrast vs. Concession — unresolved versus resolved. Same coupling, but `is_concession` flips and a resolver atom must be identified, which changes the effective strength by the 0.45 discount.

3.5 Target counts and the composition of the corpus

Phase-2 target: **600 items** in **120 families** of 5 rungs (Section 5), spanning all 36 subject topics (Section 3.2) so cross-dataset comparison stays possible. At ~ 8 gold edges per item that is $\sim 4,800$ gold relations, comparable to LOREFACT’s 5,706 logical statements. Table 9 gives the per-sense allocation — the *facet* axis, orthogonal to the topic axis above.

Sense	Coupling	Edges	Share	Cov.	Rationale
Cause-Effect	ENTAILMENT	700	14.6%	covered	most frequent in natural discourse
Effect-Cause	ENTAILMENT	450	9.4%	covered	direction-confusion partner of the above
Evidence	ENTAILMENT	400	8.3%	partial	epistemic/ontic boundary with causal
Condition	ENTAILMENT	450	9.4%	covered	confusable with Precedence (LOREFACT’s own failure mode)
Instantiation	ENTAILMENT	350	7.3%	new	no prior gold
Restatement	EQUIVALENCE	350	7.3%	new	new coupling; no prior gold
Contrast	CONTRADICTION	400	8.3%	new	as distinct from Concession
Concession (<i>unresolved</i>)	CONTRADICTION	250	5.2%	new	the undiscounted baseline for the resolved case
Concession (<i>resolved</i>)	CONTRADICTION	250	5.2%	new	exercises the 0.45 discount and the resolver heuristic
Alternative	EXCLUSIVE	400	8.3%	new	new coupling; no prior gold
Disjunction	CO_NECESSITY	350	7.3%	new	new coupling; no prior gold
Precedence	NONE	200	4.2%	none	ordering-only invariance targets
Succession	NONE	200	4.2%	none	ordering-only invariance targets
None (<i>distractors</i>)	NONE	$\geq 1,500$	—	none	separate in-window pool; the denominator for over-connection

Table 9: Per-sense gold-edge targets. **new** no prior gold data anywhere **partial** covered, but only coarsely **none** no MRF factor (NONE). The **six** new senses — Instantiation, Restatement, Contrast, Concession, Alternative, Disjunction (the last three counted once each, with Concession split here by resolution status) — are over-represented relative to their natural discourse frequency on purpose: the benchmark’s job is to grade the miner where no gold data exists. Both raw and frequency-reweighted metrics must be reported (Section 8) so natural-text performance stays estimable.

The **None** pool deserves its separate line. Our documented failure mode is *over-connection*: a pair-only judgment yields 6–9 relations per atom and invents contradictions on coherent text, which is why mining is response-grounded and there is no ungrounded path. A benchmark that contains only real relations cannot measure that failure at all. The distractors are therefore first-class data, and they must be *in-window* pairs (Section 9, R2) — a pair the candidate policy never scores is not a test of anything.

3.6 Against LoReFact’s 78%-false balance

LOREFACT makes 4,429 of 5,706 statements logically false by design, and for their task that is defensible: detection is a binary decision, and a detector is trained and measured mostly on positives. Their own Section 2.3 explains the skew as a direct consequence of instructing the generator that “each logical statement must contain a logical error”.

For a *graded, graph-level* score the same choice is disqualifying, for three reasons:

- (1) **The high end goes unvalidated.** We need to know that a coherent response scores near the top of the range. A corpus that is 78% incoherent contains almost no evidence about that region. The AeroParts family spans only 0.586–0.620, so the interesting variation is *narrow* and sits entirely inside the region such a corpus under-samples.
- (2) **A conflict detector would pass.** A degenerate scorer that reports $1 - \mathbb{I}[\text{any contradiction}]$ scores well on a mostly-incoherent corpus and is useless. Only balanced ladders separate “tracks coherence” from “detects conflicts”.
- (3) **Saturation is invisible.** A readout that saturates low cannot be distinguished from a correct one without coherent items to contrast against.

Adopted composition.

- **Relation validity: 55% valid / 45% invalid.** A *valid* relation is one the text genuinely draws and that genuinely holds; *invalid* covers wrong sense, reversed direction, a factually false endpoint, or a spurious link.
- **Rung level:** every 5-rung family contains exactly one fully coherent rung and one maximally perturbed rung, so 20% of items sit near the ceiling and 20% near the floor *by construction*. This is what makes ceiling/floor calibration — and hence the dynamic-range check of Section 8.3 — possible at all.

4 The error and perturbation taxonomy

LOREFACT has a single error axis: a logical statement is false either because the relation is wrong between correct claims, or because an endpoint claim is false. Our score is graded and graph-level, so we need an axis per mechanism that can move it.

4.1 The sense-confusability matrix

Wrong-sense perturbations should target *confusable* pairs, not random ones. The pairs in Table 10 are seeded from LOREFACT’s own error analysis (their §4.4) and from our taxonomy’s structure. Weight 3 pairs drive item sampling.

4.2 The ten operators

Table 11 defines the operator set and, for each, the expected direction of every readout: $\downarrow$ must fall, $\uparrow$ must rise, $=$ must be invariant within tolerance, $\sim$ unconstrained.

Definitions, briefly: P-WS relabels a gold edge to a confusable sense per Table 10; P-DR swaps source and target on a directed sense; P-EF rewrites a Contrast as an Alternative or the reverse; P-CR deletes the resolver atom, leaving the tension unresolved; P-IC adds one atom contradicting an existing one; P-BC removes a middle link of an entailment chain; P-OS permutes sentences without changing wording; P-SR asserts an unwarranted link between two atoms the original kept separate; P-DL removes the connective that licensed a gold edge; P-OO rewrites a Precedence as a Succession preserving the actual chronology.

Gold	Distractor	Wt	Why confusable, and what it costs
Condition	Precedence/Succession	3	LOREFACT’s documented failure: “when” read as a time marker. Coupling-changing (ENTAILMENT $\rightarrow$ NONE): the edge disappears
Cause-Effect	Effect-Cause	3	direction only; coupling preserved
Contrast	Alternative	3	exhaustiveness. Coupling-changing (CONTRADICTION $\rightarrow$ EXCLUSIVE)
Alternative	Disjunction	3	which world is excluded: (1,1) vs. (0,0). Coupling-changing
Concession	Contrast	3	resolution present or not; coupling preserved, is_concession flips
Cause-Effect	Evidence	2	ontic vs. epistemic; coupling preserved
Instantiation	Restatement	2	specificity. Coupling-changing (ENTAILMENT $\rightarrow$ EQUIVALENCE)
Cause-Effect	Condition	2	actual vs. hypothetical
Precedence	Cause-Effect	2	<i>post hoc</i> fallacy. Coupling-changing (NONE $\rightarrow$ ENTAILMENT): an edge appears where none belongs
Restatement	Cause-Effect	1	
Disjunction	None	1	

Table 10: Sense confusability. **Coupling-changing** confusions damage the Markov network — a different factor table, or none at all — while coupling-preserving ones damage only interpretability. Section 8.1 requires the two be reported separately, since they have different consequences for a downstream consumer of the score.

4.3 Two strict invariance tests

These are the sharpest instruments in the design, because the taxonomy’s own structure licenses a *falsifiable prediction of no effect* — and a no-effect prediction cannot be satisfied by accident.

Ordering-only invariance (p-oo). `COMPILE[Precedence].level1 = NONE` with `ordering_only=True`, so a `Precedence $\leftrightarrow$ Succession` edit adds **no factor to the network**. All four readouts must be identical up to miner noise; the tolerance is $\varepsilon = 2\sigma$ (Section 5.3). A failure means the miner is leaking temporal cues into truth couplings — which is precisely LOREFACT’s confirmed failure mode (“when” read as temporal), now measurable as a score-level defect rather than only a label error.

Direction invariance (p-dr). The Markov network is undirected, and `edge_factor_values` is symmetric for a given coupling. So reversing a relation’s direction must move *no* readout, while Target-A direction accuracy *must* catch it. This cleanly separates two claims we otherwise conflate: “the miner knows the direction” (an interpretability claim, which we do make) from “the score depends on the direction” (a modeling claim, which we do *not*). Any readout movement under P-DR is a bug, not a finding.

4.4 The sign-discordant operator

P-CR (remove resolution) is the single most valuable test in the suite, and it exists only because of Finding 1. Removing a resolution *raises* the effective conflict strength (no 0.45 discount), which

	Operator	Grades	<i>mean_marginal</i>	<i>consistency</i>	<i>log_partition</i>	
1	P-WS wrong sense	A	$\sim$	$\sim$	$\sim$	
2	P-DR direction reversal	A	$=$	$=$	$=$	★
3	P-EF exhaustiveness flip	A, B	$\downarrow$	$\downarrow$	$\downarrow$	
4	P-CR remove resolution	A, B	$\downarrow$	$\uparrow$	$\downarrow$	†
5	P-IC inject contradiction	B	$\downarrow$	$\downarrow$	$\downarrow$	
6	P-BC break causal chain	B	$\downarrow$	$=$	$\downarrow$	
7	P-OS shuffle order	B	$=/\downarrow$	$=$	$=$	
8	P-SR spurious relation	A, B	$\downarrow$	$\downarrow$	$\downarrow$	
9	P-DL drop relation	A	$\uparrow$	$\uparrow$	$\uparrow$	
10	P-OO ordering-only edit	A, B	$=$	$=$	$=$	★

Table 11: Perturbation operators and per-readout expected directions. “Grades” names the target from Section 8: A = per-pair relation labels, B = family ranking. ★ marks the two *strict invariance* tests (Section 4.3); † marks the sign-discordant operator (Section 4.4). Rows 9’s $\uparrow$ assumes the dropped relation was a conflict; dropping a supporting edge is $\sim$.

lowers *mean_marginal* and *log_partition* while *raising consistency*. Encoding that as a per-readout expected direction converts a known theoretical artifact into a regression test: a system that moves all three the same way under P-CR has either lost the discount or lost the distinction between belief-reading and event-reading readouts.

5 Coherence-ordered families

Target B grades the readouts by *ranking*, not by absolute value. The unit is a *family*: a set of responses differing only by controlled perturbations, whose correct coherence order is known by construction.

5.1 Scope: three readouts

LoCoBench grades **three** readouts:

What is excluded, and why it is not an inconsistency. `LCS_METHODS` in `lcs_scorer.py` offers a fourth readout, **reified** — $P(R=1)$ for an added coherence node (Eqs. 6–7) — and the scorer will continue to compute it. It is out of *benchmark* scope for three reasons, none of which is a defect in the readout:

- (1) **It is not independent of consistency.** Both read conflict *activity* rather than belief, so both dip at the concession rung for the same structural reason (Finding 1). Grading both would double-count one signal in every C2 contract and every quirk-conformity figure.
- (2) **It carries a tunable constant.** The reified node needs a Bernoulli prior (`reified_prior`, default 0.5), so its value is not constant-free the way the three in-scope readouts are. A benchmark number that moves with an unreported hyper-parameter is not comparable across systems.

Readout	Reads	Why it is in scope
<code>mean_marginal</code>	belief	the shipped default headline (deep-dive Eq. 4): MRF-native, monotone, constant-free, in $[0, 1]$, read straight off Merlin’s MAR marginals
<code>consistency</code>	event activity	$P(\text{no conflict edge jointly active})$ (Eq. 5). The only in-scope readout that reads conflict <i>activity</i> rather than belief, which is what makes the concession quirk of Finding 1 observable at all
<code>log_partition</code>	mass	the normalized $\log Z$ readout (Eq. 8), graded in $[0, 1]$ between a contradiction-free ceiling and a saturated floor

Table 12: The three readouts LoCoBench grades. Each reads the same network a different way — belief, event activity, mass — so the set spans the distinction Finding 1 turns on while staying small enough that per-readout ordering contracts remain legible.

(3) **It costs inference without adding a distinction.** Dropping it removes the R-node MAR run from the per-item budget while leaving all four *kinds* of Target-B claim — full chain, quirk chain, endpoints, invariance — fully testable.

So the benchmark grades a strict subset of what the code computes. A Phase-2 harness may record `reified` alongside the three as a diagnostic; it simply does not enter any metric, contract or acceptance threshold in this document.

5.2 The five-rung ladder

The rung structure reuses the deep-dive’s own variant family (its Table “variants”), whose exact values — brute-forced over 2^{16} worlds — give us a calibrated reference ladder (Table 13).

Rung	Name	Perturbation relative to base	ref. LCS	ref. $\log Z$
0	<code>worse</code>	+ a spurious contradiction (P-SR)	0.586	−10.54
1	<code>base</code>	as generated; all planned conflicts present	0.607	−9.64
2	<code>concession_resolved</code>	resolver present; conflict discounted $.80 \rightarrow .55$	0.612	−9.64
3	<code>fix_one_conflict</code>	drop one unresolved conflict (P-DL)	0.613	−8.95
4	<code>coherent</code>	all conflicts removed	0.620	−8.25

Table 13: The reference ladder, with exact `mean_marginal` and $\log Z$ from the AeroParts model. Note how narrow the band is (0.586–0.620, a range of 0.034): the margin rule of Section 5.3 exists because adjacent-rung deltas are of the same order as miner noise.

5.3 Ordering contracts, per readout

Per Finding 1 the three readouts do *not* move together across the rungs, so the ordering contract is stated per readout. Three classes:

C1 — full chain (`mean_marginal`, `log_partition`): $\rho(r_0) < \rho(r_1) < \rho(r_2) < \rho(r_3) < \rho(r_4)$, strict, all four adjacent pairs.

C2 — quirk chain (`consistency`): require $\rho(r_0) < \rho(r_1)$ and $\rho(r_3) < \rho(r_4)$; **assert the inversion** $\rho(r_2) \leq \rho(r_1)$ as a positive prediction; leave $r_2 \rightarrow r_3$ unconstrained.

C3 — endpoints (all three readouts): $\rho(r_0) < \rho(r_4)$, strict. The weakest and most robust claim; it survives every quirk, and is therefore the headline ranking number.

The margin rule. Adjacent-rung deltas are small (Table 13), so a constraint is *scored* only when $|\Delta| > 2\sigma$, where σ is the readout’s run-to-run standard deviation measured over 5 re-mines of the same rung. Otherwise the constraint is marked **indeterminate** and excluded from the denominator.

This creates an obvious gaming channel that the metrics must close: *a sufficiently noisy system marks everything indeterminate and scores a vacuous 100%*. Section 8.3 therefore requires the indeterminate rate be reported alongside every ranking number, and the two read together.

5.4 Family types

Type	Focus	n	Rung structure
F-CONF	conflicts	40	the reference ladder of Table 13
F-CHAIN	entailment chains	25	P-BC at decreasing depth: break 3 links, 2, 1, 0, then a coherent rewrite
F-EXH	exhaustiveness	20	P-EF walks Alternative $\rightarrow$ Contrast $\rightarrow$ resolved; tests whether EXCLUSIVE is scored strictly below CONTRADICTION
F-ORDER	order sensitivity	20	P-OS at decreasing severity (full shuffle, block, adjacent swap, original) <i>plus</i> one P-OO rung that must be flat
F-INV	invariance only	15	all five rungs are P-OO/P-DR edits, so the required pattern is a flat line

Table 14: Family types. F-INV is the control: any monotone trend there is a false positive, and it guards against a score that merely correlates with edit count rather than with coherence. F-ORDER is dual-purpose, carrying four ordering rungs and one invariance rung in the same family.

5.5 The seed split

The nine fixtures in `data/lcs/` become the hand-authored **dev** split, with the ordering constraints `research_plan.md` §9 already asserts encoded as degenerate two-rung families:

- `example-2-biography > example-2-biography-contradicted;`
- `example-5-renda-K > example-5-renda-S;`
- `aeroparts-recall` as the reference five-rung family, with the published exact values as `reference_values` (Section 7).

They are dev rather than test for a specific reason given in Section 7.3: the gold edges recoverable from the existing builder are *lossy in sense granularity*, so they can only grade coupling, not sense.

6 The prompt suite

Data generation is five prompts (P1–P5); admission to the benchmark is four more (v1–v4) plus an annotation protocol (v5). This section is the map: Figure 1 shows how they compose, Table 15 gives each prompt’s contract, and **the full texts are in Appendix A**, one subsection per prompt, copy-ready.

The suite keeps LOREFACT’s house style throughout — numbered instructions, few-shot exemplars, bracketed output tags, a **Your Task:** trailer — so the lineage is explicit and the diffs against their Tables 11–14 are legible. The validation prompts keep their function-call idiom from Tables 20–25, which has the useful side effect of making their input/output contract self-declaring. Sense menus match `lcs/prompts.py::SENSE_MENU` exactly.

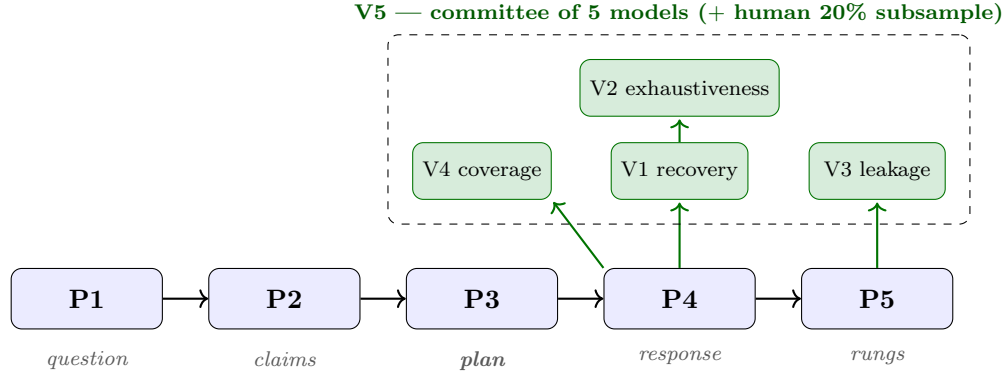


Figure 1: The generation pipeline (blue, left to right) and the admission gates (green, above). P1–P5 produce data; v1–v4 decide whether an item is admitted, and v5 is the committee that runs them. v1 is the load-bearing gate: it re-derives the relation graph from the response *without seeing the plan*, which is what proves the gold is recoverable from the text rather than merely asserted about it. v2 runs on v1’s conflict edges only, adjudicating the exhaustiveness boundary that separates CONTRADICTION from EXCLUSIVE.

ID	Role	Inputs	Output	Gate
<i>Generation — these produce the data</i>				
P1	topic $\rightarrow$ question	TOPIC	one bracketed question	—
P2	atomic claims	QUESTION	24 tagged claims + 2 holdings	—
p3	relation plan	QUESTION, CLAIMS	JSON graph: atoms, relations, non-relations	schema-valid; ≥ 1 each of Alternative, Disjunction, Restatement, Concession, Precedence/Succession
P4	response	QUESTION, PLAN	prose response, ≥ 500 words	v1, v3, v4
P5	perturbation	RESPONSE, PLAN, OPERATOR	perturbed response + JSON diff	one operator applied; length within 15%
<i>Validation — these decide what is admitted</i>				
v1	relation recovery	RESPONSE, ATOMS	recovered relation list	$\geq 80\%$ coupling , $\geq 70\%$ sense , else drop the edge or regenerate
v2	exhaustiveness	CLAIM_A, CLAIM_B, RESPONSE	one of A/B/C/D	unanimity for an EXCLUSIVE label (Section 9, R1)
v3	naturalness, leakage	RESPONSE	3 scores + 3 span lists	all scores ≥ 4 ; leakage and hedging empty
v4	atom coverage	RESPONSE, ATOMS	per-atom status + position	100% asserted ; positions re-verify the window
v5	annotation protocol	—	—	<i>not a prompt</i> : the committee and agreement thresholds (Section 6.2)

Table 15: **The prompt suite index.** Full texts in Appendix A. Every NAME above is a [NAME.PLACEHOLDER] in the prompt body. Note v5 carries no prompt — it is the protocol that runs v1–v4, kept in this series because it is what makes their verdicts binding.

6.1 Why P3 replaces LoReFact’s logical-statement stage

The architectural departure is P3. LOREFACT generates prose “logical statements” and annotates them afterwards; we generate a *machine-readable relation graph* first and realize it in prose second. So the gold labels are an **input** to generation rather than an output of annotation, which removes a whole class of annotation error — at the cost of requiring v1 to prove the graph is actually recoverable from the text.

That trade is worth making here for a reason specific to our taxonomy. Six of our thirteen senses have no gold data anywhere (Table 3), so post-hoc annotation would be asking annotators to find relation facets they have never seen labelled, at whatever rate those facets happen to occur in generated prose. Planning the graph first makes coverage of the rare facets a *generation constraint* (P3 instruction 5) rather than a matter of luck.

It also changes what LOREFACT’s 78%-false skew is for. Their skew follows from instructing the generator that every logical statement must contain an error; ours is a **validity** field with an explicit 55/45 target (P3 instruction 8), so the balance is a dial rather than a consequence.

6.2 V5 — the annotation protocol

LOREFACT used three graduate-student annotators with a third-annotator tiebreak (reaching 92.18% agreement) plus a five-model committee with unweighted majority voting for logical factuality. Ours mirrors that shape with two additions forced by our design.

Model committee (all items). Five models drawn from at least three distinct families, each running V1, V2 and V4 independently at temperature 0; per-edge label by majority, ties escalated to a human. **Generation/validation disjointness:** the model that ran P3/P4 for an item is *excluded* from that item’s committee, per R3.

Human annotation (stratified 20%). Three annotators on 120 items, stratified to cover *all* EXCLUSIVE and CO_NECESSITY gold edges, all resolved Concessions, all P-OO invariance rungs, and a random remainder. These are exactly the facets with no prior art and the highest expected model unreliability, so uniform sampling would waste the human budget.

Statistics and thresholds. Fleiss’ κ on the 13-way sense and 6-way coupling labels (human-only and committee-only); Cohen’s κ on the binary exhaustiveness decision, reported *separately* as the expected weak point; Krippendorff’s α (ordinal) on strength bands; and human-versus-committee agreement, which is what licenses the committee as a human proxy on the other 80%. Acceptance: $\kappa \geq 0.70$ on coupling, ≥ 0.60 on sense, ≥ 0.55 on exhaustiveness. A facet below threshold ships marked **low_agreement** and is excluded from headline metrics but reported as a finding — an unreliable facet is itself a result about the taxonomy.

7 The gold schema

7.1 A benchmark item

The item format is a **strict superset** of `data/lcs/*.json`, so the nine shipped fixtures remain valid and `RelationMiner.mine_from_atoms` loads a benchmark item unchanged. Existing keys (`id`, `name`, `source`, `response`, `num_atoms`, `atoms`, `notes`) keep their meaning; four blocks are added.

```

{
  "id": "locobench-f012-r1",
  "name": "Aviation component failure - base",
  "source": "generated:P4/<model>@<date>",
  "response": "<full response text>",
  "num_atoms": 16,
  "atoms": [
    {
      "id": "a0", "text": "AeroParts supplied turbine blades to the carrier.",
      "label": "a1", "factual": true, "role": "claim",
      "sentence_index": 1, "char_span": [0, 52]}
  ],
  "notes": "...",

  "relations": [
    {
      "id": "r003",
      "source_id": "a10", "target_id": "a11",
      "level2_sense": "Alternative",
      "level1_coupling": "exclusive",
      "directed": false,
      "exhaustive": true,
      "ordering_only": false,
      "intended_strength_band": "strong",
      "strength_range": [0.85, 1.00],
      "validity": "valid",
      "error_kind": null,
      "is_concession": false,
      "is_resolved_concession": false,
      "resolver_atom_id": null,
      "position_distance": 1,
      "in_candidate_window": true,
      "window_admission": "window",
      "provenance": {
        "planned_by": "P3/<model>",
        "recovered_by": ["<m1>", "<m2>", "<m3>"],
        "recovery_rate": 0.80,
        "exhaustiveness_verdict": "A",
        "exhaustiveness_votes": {"A": 4, "B": 1, "C": 0, "D": 0},
        "human_confirmed": true,
        "annotator_agreement": 1.0,
        "low_agreement": false
      }
    }
  ],

  "non_relations": [
    {
      "source_id": "a1", "target_id": "a12", "position_distance": 3,
      "in_candidate_window": true,
      "provenance": {"recovered_as_none_by": 5, "human_confirmed": true}}
  ],

  "expected": {
    "family_id": "f012",
    "family_type": "F-CONF",

```

```

    "rung_index": 1,
    "rung_name": "base",
    "perturbation": {"operator": null, "target": null, "parent_rung": null},
    "readout_directions": {
        "mean_marginal": null, "consistency": null, "log_partition": null
    },
    "invariants": [],
    "reference_values": {
        "mean_marginal": 0.607, "consistency": null, "log_partition": -9.64
    }
},

"meta": {
    "domain": "Engineering & Technology",
    "topic": "aviation component failure investigation",
    "split": "test",
    "word_count": 623,
    "validation": {
        "v1_coupling_recall": 0.88, "v1_sense_recall": 0.75,
        "v3_fluency": 5, "v3_leakage": [], "v4_coverage": 1.0
    }
}
}

```

Field notes, in decreasing order of consequence:

`level1_coupling` must satisfy `COMPILE[level2_sense].level1 == level1_coupling`. A builder assertion **must** enforce this, so gold can never silently contradict `taxonomy.py` (Table 8). The same applies to `directed` and `ordering_only`, which mirror `SenseSpec`.

`exhaustive` is meaningful only for CONTRADICTION/EXCLUSIVE edges, where `true` $\Leftrightarrow$ Alternative. It is stored *explicitly* rather than derived from the sense: that redundancy is deliberate, because it is what lets a disputed edge be flagged `low_agreement` while still carrying a sense, and it is the field the exhaustiveness metric grades against.

`in_candidate_window` / `window_admission` (`window` | `gate` | `discourse_promoted` | `out_of_window`) are computed at build time by actually running `candidate_pairs.select(policy="gated", window=4)` and recording the verdict. This makes gold recoverability under the shipped pair policy an explicit, queryable property rather than an assumption (R2).

`readout_directions` carries the per-readout expectation from Table 11 for the perturbation that produced this rung — one of `decrease`, `increase`, `invariant`, `unconstrained` per readout, relative to `parent_rung`. It is null on a base rung. This is where Finding 1 lives in the data.

`role` on an atom is `claim` | `holding` | `distractor`, so a resolver can be located without re-running the heuristic.

7.2 The family manifest

```

{
    "manifest_version": "1.0",
    "families": [
        {
            "family_id": "f012",
            "family_type": "F-CONF",
            "domain": "Engineering & Technology",

```

```

"topic": "aviation component failure investigation",
"rungs": [
  {"index": 0, "name": "worse", "item_id": "locobench-f012-r0",
   "operator": "spurious_relation", "parent": 1},
  {"index": 1, "name": "base", "item_id": "locobench-f012-r1",
   "operator": null, "parent": null},
  {"index": 2, "name": "concession_resolved", "item_id": "...-r2",
   "operator": "add_resolution", "parent": 1},
  {"index": 3, "name": "fix_one_conflict", "item_id": "...-r3",
   "operator": "drop_relation", "parent": 2},
  {"index": 4, "name": "coherent", "item_id": "...-r4",
   "operator": "drop_relation", "parent": 3}
],
"ordering_constraints": [
  {"id": "c1", "class": "C1",
   "readouts": ["mean_marginal", "log_partition"],
   "chain": [0, 1, 2, 3, 4], "strict": true},
  {"id": "c2", "class": "C2",
   "readouts": ["consistency"],
   "required": [[0, 1], [3, 4]],
   "expected_inversions": [[1, 2]],
   "unconstrained": [[2, 3]]},
  {"id": "c3", "class": "C3",
   "readouts": ["mean_marginal", "consistency", "log_partition"],
   "required": [[0, 4]], "strict": true}
],
"invariance_constraints": [],
"noise_sigma": {"mean_marginal": 0.004, "consistency": 0.011,
                 "log_partition": 0.06},
"margin_rule": "abs_delta > 2 * noise_sigma else indeterminate"
}
],
"legacy_pairs": [
  {"family_id": "seed-renda",
   "rungs": [{"index": 0, "item_id": "example-5-renda-S"},
             {"index": 1, "item_id": "example-5-renda-K"}],
   "ordering_constraints": [{"id": "c3", "class": "C3",
                             "readouts": ["mean_marginal"], "required": [[0, 1]], "strict": true}]},
  {"family_id": "seed-biography",
   "rungs": [{"index": 0, "item_id": "example-2-biography-contradicted"},
             {"index": 1, "item_id": "example-2-biography"}],
   "ordering_constraints": [{"id": "c3", "class": "C3",
                             "readouts": ["mean_marginal"], "required": [[0, 1]], "strict": true}}}
]
}

```

expected_inversions in the c2 constraint is Finding 1 made executable: the harness *asserts* that consistency falls from rung 1 to rung 2, rather than exempting the step.

7.3 Migrating the seed fixtures

scripts/build_lcs_examples.py already contains hand-authored gold relations, as 4-tuples (src_label, trg_label, level1.type, relation_class) with relation_class ∈

{prerequisite, invalidation} — and `_build_record` discards them. The comment at its definition promises they are “expanded to a0-based ids at build time”; that expansion was never implemented, so the data is live only inside the script. Phase 1 specifies surfacing it:

- `prerequisite` → `level2_sense: "Cause-Effect", level1_coupling: "entailment", directed: true;`
- `invalidation` → `"Contrast", CONTRADICTION, directed: false;`
- the AeroParts equivalence edge → `"Restatement", EQUIVALENCE;`
- the AeroParts blame conflict additionally gets `is_concession: true, is_resolved_concession: true` and its `resolver_atom_id`, per the fixture’s existing notes.

Why this makes the seed set dev, not test. The migration is **lossy in sense granularity**: `prerequisite` covers whatever the true sense was, so mapping it to Cause-Effect is a guess wherever the truth is Evidence, Condition or Instantiation. Migrated edges are therefore graded on **coupling only** unless a human re-annotates the sense. Migrated edges carry `provenance.planned_by: "hand-authored"` and must still pass V1 to be scored.

Two housekeeping items to resolve before Phase 2 touches that script. `data/lcs/example-6-incident.json` is **orphaned**: the builder registers only eight examples and does not know it, and `data/lcs/README.md` omits it. Separately, that README has been **hand-edited since generation** — it documents the two-stage `node_priors` path that `_write_readme` does not emit — so re-running the builder would regress it. Either fold example 6 into the builder and re-sync the README generator, or move both under the new benchmark builder.

8 Metrics

Notation: item k has gold edge set G_k and candidate-pair set C_k ; a system produces predicted edges P_k . Let G^{edge} be the gold edges with a non-NONE coupling, and N the `non_relations` pool. Write c, s, d for gold coupling/sense/direction and $\hat{c}, \hat{s}, \hat{d}$ for predictions.

All Target-A metrics are computed over candidate pairs only. A pair the policy never scored cannot be a miner error. Gold edges with `in_candidate_window: false` are excluded from recall and reported separately as the **structural-miss rate** — conflating the two would blame the miner for the gate’s decisions (R2).

8.1 Target A — the miner as a classifier

A1. Coupling accuracy. $\text{CouplingAcc} = |\{e \in G^{\text{edge}} : \hat{c}(e) = c(e)\}| / |G^{\text{edge}}|$.

A2. Sense accuracy. Same denominator with $\hat{s}, s$. Report also **SenseAcc-hard**, restricted to edges whose confusability weight is 3 (Table 10), and split every figure into *coupling-changing* versus *coupling-preserving* confusions.

A3. Direction accuracy. Over gold edges where `COMPILE[s].directed` holds *and* the coupling was predicted correctly (direction on a wrong coupling is not meaningful):

$$\text{DirAcc} = \frac{|\{e : \hat{d}(e) = d(e) \wedge \hat{c}(e) = c(e)\}|}{|\{e \in G : \text{directed}(s(e))\}|}.$$

A4. None-detection P/R/F1. With P_{none} the pairs the system labels NONE:

$$\mathcal{P} = \frac{|N \cap P_{\text{none}}|}{|P_{\text{none}}|}, \quad \mathcal{R} = \frac{|N \cap P_{\text{none}}|}{|N|}, \quad F_1 = \frac{2\mathcal{P}\mathcal{R}}{\mathcal{P} + \mathcal{R}}.$$

Note $1 - \mathcal{P}$ is the over-connection rate — our documented failure mode, and the reason the None pool is first-class data (Section 3.5).

A5. Exhaustiveness accuracy. Over gold edges with coupling $\in \{\text{CONTRADICTION, EXCLUSIVE}\}$, the rate at which **exhaustive** is predicted correctly, *plus* the explicit 2×2 Contrast/Alternative confusion matrix. This metric has no LOREFACT analogue and is the one that justifies EXCLUSIVE as a separate coupling.

A6. Concession-resolution accuracy. Over gold **is_concession** edges: the binary “did the system mark it resolved”, reported separately from exact **resolver_atom_id** match.

A7. Strength-band agreement. Exact-match rate; *adjacent* accuracy (off-by-one band counted correct); Krippendorff’s α (ordinal); and the mean absolute deviation of the predicted p from the nearest edge of the gold band, zero inside the band.

A8. Confusion matrices. 13×13 over senses and 6×6 over couplings, each with an extra **structural** row/column for out-of-window gold. Published as the primary diagnostic artifact — the aggregate numbers above are summaries of it.

8.2 LoReFact-comparable analogues

LOREFACT’s three headline logic metrics, lifted to a graded, graph-level setting (Table 16). The point of this table is comparability: a reader of both papers should be able to line the numbers up.

LoReFact	LoCoBench	Definition and difference
LogicRecall	EdgeRecall	$ \{e \in G^{\text{edge}} : \hat{c}(e) \neq \text{NONE}\} / G^{\text{edge}} $ — fraction of in-window gold edges detected as <i>some</i> relation. Complements $\mathcal{R}$ above
LogicTypeAcc	CouplingAcc, SenseAcc	their 5-class becomes our 6-class coupling and 13-class sense, i.e. two granularities instead of one
LogicFactualityAcc	ValidityAUROC, BandMAE	their binary “does the dependency hold” becomes graded: use the miner’s edge probability p as a score for validity and report AUROC. A thresholded ValidityAcc at $p=0.5$ is also reported for direct numeric comparison
—	CoherenceAUC	new: pooling all items, each readout used as a score for “this rung is in the top half of its family”. One number per readout, comparable across readouts <i>and</i> against DiscoScore / G-Eval / SelfCheckGPT-NLI

Table 16: LOREFACT metric analogues. The graded-versus-binary shift in the third row is the substantive one: a binary logic-factuality label discards exactly the information our relation probability carries.

8.3 Target B — readout ranking

For readout ρ and family f with rungs $r_0..r_4$, and σ_ρ the measured run-to-run noise:

B1. Pairwise ranking accuracy, over the constraint’s required pairs, excluding indeterminate ones:

$$\text{PRA}(\rho) = \frac{\sum_f |\{(i, j) \text{ required} : \rho(r_j) - \rho(r_i) > 2\sigma_\rho\}|}{\sum_f |\{(i, j) \text{ required} : \text{determinate}\}|}.$$

Must be reported with the indeterminate rate. A system can score $\text{PRA} = 1$ trivially by being noisy enough that every pair is indeterminate; the two numbers are only meaningful together.

B2. Kendall τ -b between the readout’s rung ordering and the gold rung index, per family, averaged. Computed on the full C1 chain for `mean_marginal` and `log_partition`; on the C2-required sub-chain only for `consistency` since τ over the full chain is not meaningful given the predicted inversion.

B3. Monotonicity violation rate. Fraction of C1 adjacent pairs moving the wrong way by more than $2\sigma_\rho$. Strict violations only — noise-level wobble is not a violation.

B4. Quirk-conformity rate (C2, `consistency` only). Fraction of families where the readout *does* invert at rung $1 \rightarrow 2$ as Finding 1 predicts. **Target is high, not low:** a system monotone here is not implementing the concession discount.

B5. Invariance pass rate. Over P-OO and P-DR rungs, $\text{INV}(\rho) = |\{|\rho(\text{perturbed}) - \rho(\text{parent})| \leq 2\sigma_\rho\}|/|\text{items}|$, reported separately as INV-ORDERING and INV-DIRECTION. This is the one metric where all three readouts share the same strict expectation; a INV-DIRECTION failure means the factor tables have become direction-sensitive.

B6. Endpoint separation and dynamic range. $\rho(r_4) - \rho(r_0)$, mean and distribution, plus the 5th/95th percentile of ρ over all items. The reference family spans only 0.586–0.620, so a *compressed* range is expected and must be quantified rather than treated as a defect.

B7. Human correlation. Spearman ρ between each readout and human coherence ratings on the 120-item human subsample, against DiscoScore [DiscoScore], the G-Eval coherence dimension [G-Eval] and SelfCheckGPT-NLI [SelfCheckGPT], per `research_plan.md` §9.

Metric	mean_marginal	consistency	log_partition
PRA, C1 chain	✓	—	✓
PRA, C2 required	✓	✓	✓
Kendall τ	full	sub-chain	full
Monotonicity violation	✓	—	✓
Quirk conformity	—	✓	—
Invariance (P-OO, P-DR)	✓	✓	✓
Endpoint separation (C3)	✓	✓	✓
Human Spearman	✓	✓	✓

Table 17: Metric applicability by readout, following from the C1/C2/C3 contract classes of Section 5.3. Target-A metrics are deliberately absent: they grade the miner upstream of scoring, so a miner regression is diagnosable without re-running inference.

9 Risks and Phase-2 open questions

R1 — LLM-generated exhaustiveness may be unreliable (highest risk). Deciding Contrast versus Alternative requires quantifying over possible worlds, which models handle poorly, and it is exactly the boundary where EXCLUSIVE earns its existence as a separate coupling. Mitigations: V2’s explicit “coherent third state” procedure; **unanimity** (not majority) of the committee required for an EXCLUSIVE gold label; 100% human coverage of those edges in the subsample; **low_agreement** flagging. *Fallback:* seed Alternative edges from **structurally guaranteed** pairs — numeric or polarity complements (“no one was harmed” / “three people died”), where exhaustiveness follows from the semantics of negation rather than from world knowledge. *Open:* if human κ on exhaustiveness lands below 0.55, is EXCLUSIVE a defensible benchmark facet at all, or should it ship as a research-only subset?

R2 — the candidate window versus gold recoverability. Under `policy="gated"`, `window=4`, a gold edge whose endpoints are more than 4 positions apart and which fails both the similarity gate and the discourse-adjacency check is **unmineable by construction**. Handled in three places: P3 instruction 4 biases generation into the admissible set; build time records `in_candidate_window / window_admission` per edge by running the real selector; the metrics exclude out-of-window gold and report the structural-miss rate separately. *Open:* P3 constrains *planned* positions but P4’s realized order may drift, so V4’s `position` output must re-verify the constraint post hoc and items that break it must regenerate. *Also open:* should Phase 2 ship a deliberate **long_range** subset of out-of-window gold to grade *the gate itself* as a retrieval problem (precision/recall of admission)? That is a different and probably valuable evaluation, and the schema already supports it.

R3 — self-generation bias. If the same model family generates (P3/P4) and mines (Prompt A/B), the miner may recover its own lexical fingerprints rather than genuine discourse structure, inflating every Target-A number. Mitigations: per-item generator exclusion from the validation committee (Section 6.2); the generator recorded in `source` and stratified across ≥ 3 families so per-generator breakdowns are computable; and a **cross-generation control** — Target-A reported both pooled and restricted to items whose generator family differs from the miner’s, with the gap published as a bias estimate. *Open (recommended yes):* a small human-authored slice, say 50 items with no LLM generation at any stage, as an unbiased anchor. It is cheap relative to its diagnostic value, and the nine existing fixtures are a precedent for the format.

R4 — Restatement’s strength prior. Per Finding 2 (Section 2.2) the 0.90 prior is unreachable on the production path, so it cannot be tested through the pipeline. What Phase 2 *can* report is the distribution of mined strength on gold Restatement edges — i.e. whether 0.90 is empirically justified. *Open:* if mined Restatement strength centres well away from 0.90, the prior should be re-estimated or removed rather than left as unreachable code.

R5 — the MLN formulation cannot be benchmarked end to end. `formulation="mln"` constructs but raises at `score()`, and EXCLUSIVE / CO_NECESSITY raise in that path specifically. Target B is therefore scoped to the MRF. The MLN’s closed-form pairwise fragment is verified to reproduce the MRF exactly, so nothing is lost for the pairwise facets; beyond-pairwise structure is out of scope until an engine exists.

R6 — generation cost. LOREFACT reports ≈ 110 LLM calls and $\approx \$0.32$ per sample for their *evaluation* pipeline alone. Ours adds a 5-model validation committee over V1/V2/V4 and a 5-rung ladder per family. Phase 2 should cost the run before committing to 600 items, and the schema’s `reference_values` exist partly so that cheap dry-run scoring (the brute-force oracle, and `score_all`’s shared base inference) can substitute during harness development.

On the inference side the three-readout scope (Section 5.1) costs **5 Merlin calls per item**, not 6: a base MAR and a base PR shared by all three, plus `consistency`’s U-chain MAR and `log_partition`’s contradiction-free ceiling PR and saturated-floor MAP. Excluding `reified` removes exactly its R-node MAR run. Over 600 items $\times$ 5 rungs that is 3,000 items and $\approx 15,000$ Merlin invocations, so the saving is real but second-order next to the LLM committee cost above.

10 Summary of Phase-1 decisions

1. **Two gradable targets**, not one: per-pair relation labels (A) grade the miner as a classifier; coherence-ordered families (B) grade the three readouts by ranking. A alone would leave the score untested; B alone would give no diagnosis of *which* facets fail.
2. **Thirteen senses, five couplings**, exactly as `taxonomy.py` defines them, with EQUIVALENCE, EXCLUSIVE and CO-NECESSITY over-represented because no prior gold data exists for them.
3. **Two orthogonal organizations, named apart.** *Facets* are LoCoBench’s own structural axes (relation, item, annotation); *topic* and *domain* stay reserved for the 36 subjects inherited from LOREFACT, **all of which must be represented**. Both are coverage requirements, and both are reported — a topic breakdown says whether the benchmark generalizes across subject matter, a facet breakdown says which structural phenomena a scorer handles.
4. **Validity-balanced, not false-skewed:** 55/45 valid/invalid with a coherent and a maximally-perturbed rung in every family, departing deliberately from LOREFACT’s 78% because a graded score must be validated at both ends.
5. **A relation plan replaces prose logical statements:** gold is an *input* to generation, gated by a round-trip recovery check rather than produced by post-hoc annotation.
6. **Per-readout ordering contracts** (C1/C2/C3), because one of the three readouts provably invert at the concession rung and a global monotonicity requirement would fail a correct implementation.
7. **Two strict invariance tests** — ordering-only and direction-reversal — which are the design’s sharpest instruments because a prediction of *no effect* cannot be satisfied by accident.
8. **A superset schema**, so the nine shipped fixtures become the dev split with no format fork, graded on coupling only where their migration is lossy.

Phase 2 is the generation run: topic selection, the live prompt pipeline, the validation committee, and the harness that turns `expected` blocks into assertions.

References

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- [Li] J. Li and D. Jurafsky. *Neural Net Models of Open-Domain Discourse Coherence*. EMNLP 2017.
- [SAFE] J. Wei et al. *Long-Form Factuality in Large Language Models*. NeurIPS 2024.

A The prompts, verbatim

The nine prompts of Table 15, copy-ready. Each is preceded by its contract: the placeholders it takes, the output it must produce, and the gate its output must pass. Generation prompts (P1–P5) come first, then validation (v1–v4). v5 is not a prompt and is specified in Section 6.2.

A.1 P1 — topic to question

P1 — topic to question

Inputs: [TOPIC_PLACEHOLDER] — one of the 36 subject topics of Table 6. The prompt is run per topic, so *topic selection is not the prompt’s job*: instruction 4’s preference for adjudicated framings applies *within* whatever topic it is given, and never substitutes for covering all 36 (Section 3.2).

Output: One question, wrapped in square brackets, no preamble.

Gate: None. A weak question is caught downstream by P3 failing instruction 5.

Instructions:

1. You are given a TOPIC. Generate one open-ended QUESTION about that topic.
2. The QUESTION must begin with Why, How, What, Does, or Which.
3. The QUESTION must elicit an answer containing MULTIPLE, DIVERSE logical relationships. Aim for an answer that would naturally contain at least four of the following: a causal chain; a condition; a temporal ordering; an evidential appeal; a general claim with a specific instance; a restatement or definitional equivalence; a non-exhaustive contrast; an EXHAUSTIVE pair of competing alternatives of which EXACTLY ONE can hold; a disjunctive pair of which AT LEAST ONE must hold; a tension that an authoritative finding later resolves.
4. Prefer topics where competing explanations are adjudicated: incident investigations, litigation, diagnostic reasoning, attribution disputes, competing scientific hypotheses. Such topics naturally produce exhaustive alternatives and resolving holdings.
5. The QUESTION must be answerable in 500–900 words of formal prose.
6. The QUESTION must not itself contain the words "cause", "condition", "alternative", or "contrast", so the answer’s relations are not primed lexically.
7. Do NOT ask a multi-part question joined by "and" – one question only.
8. Wrap the QUESTION in square brackets.

9. Output the bracketed QUESTION only, with no preamble.

Use the following examples to better understand your task.

TOPIC: aviation component failure investigation

QUESTION: [How do investigators determine responsibility when a component supplied by a third party fails in service and the operator and the manufacturer offer competing explanations?]

TOPIC: antibiotic resistance in hospital settings

QUESTION: [Why does resistance emerge in some hospital units but not others, and how is a specific outbreak traced to its origin?]

Your Task:

TOPIC: [TOPIC_PLACEHOLDER]

QUESTION:

Instructions 3 and 4 are the substantive extension of LOREFACT’s Table 11, which asks only for “causal, conditional, or temporal reasoning”. Instruction 4 encodes an empirical bet: adjudicated domains are where exhaustive alternatives and resolving holdings occur naturally, so steering topic selection is cheaper than forcing the structure later. Instruction 6 prevents lexical priming that would make the miner’s job artificially easy.

A.2 P2 — atomic claim generation

P2 — atomic claim generation

Inputs: [QUESTION_PLACEHOLDER].

Output: A markdown code block: 24 claims as – bullets, each with one of the 9 tags of Table 5, plus 2 [holding] claims.

Gate: Composition must match instruction 3 exactly (14 correct, 4 incorrect, and the alt/disj/equiv pairs); the holdings must use adjudicating vocabulary.

Instructions:

1. You are given a QUESTION. Generate a set of ATOMIC CLAIMS that a well-informed answer to it might contain.
2. An atomic claim is a sentence containing a singular piece of information, which should be basic, indivisible, and independently verifiable.
3. Generate exactly 24 atomic claims, in the following required composition:
 - (a) 14 factually correct claims;
 - (b) 4 factually incorrect claims (plausible but false);
 - (c) 2 EXHAUSTIVE ALTERNATIVE claims: a pair X, Y such that exactly one of X and Y must be true – they cannot both hold, and they cannot both fail. Example pair: "No one was harmed in the incident." / "Three people died in the incident."
 - (d) 2 DISJUNCTIVE claims: a pair X, Y such that at least one must be true, but both may be true simultaneously. Example: two independent checks, at least one of which flagged a defect.
 - (e) 2 EQUIVALENT claims: a pair X, Y asserting the same fact in different words, so that either one being true makes the other true.
4. Additionally generate 2 HOLDING claims: sentences in which an authority (a court, tribunal, regulator, final report, or review board) reaches a determination that settles a dispute between two other claims in your list. A holding claim must use adjudicating language – "held", "ruled",

- "concluded", "determined", "found that", "ultimately".
5. Each claim must be a single declarative sentence with no connective linking it to another claim. Do NOT use "because", "although", "therefore", "if", "while", "however". Relations are added at a later stage, not here.
 6. Do NOT indicate which claims are factually incorrect in their wording; do NOT hedge with "assume", "might", "possibly", or "allegedly".
 7. Output as a markdown code block: an unordered list with "-" bullets. Append a tag in square brackets to every claim, one of: [correct], [incorrect], [alt-pair-1], [alt-pair-2], [disj-pair-1], [disj-pair-2], [equiv-pair-1], [equiv-pair-2], [holding].

Use the following example to better understand your task.

QUESTION: [How do investigators determine responsibility when a component supplied by a third party fails in service and the operator and the manufacturer offer competing explanations?]

ATOMIC CLAIMS:

```
'''
- AeroParts supplied turbine blades to the carrier. [correct]
- The turbine blades were installed in the carrier's A-320 fleet. [correct]
- Seventeen turbine blades failed in flight during service. [correct]
- The regulator opened a safety probe into the blade failures. [correct]
- Maintenance logs recorded abnormal vibration on the failed blades. [correct]
- The airline argued that the failures were caused by pilot error. [correct]
- The final report concluded the root cause was a metallurgical defect. [correct]
- The regulator issued an order grounding the fleet. [correct]
- Every affected aircraft was pulled from service. [equiv-pair-1]
- The regulator's order removed all affected aircraft from operation. [equiv-pair-2]
- The blades were certified under a 1998 airworthiness standard. [incorrect]
- No one was harmed in the incidents. [alt-pair-1]
- Three passengers died in one of the failures. [alt-pair-2]
- The vibration analysis flagged the defect. [disj-pair-1]
- The metallurgical assay flagged the defect. [disj-pair-2]
- The tribunal held that AeroParts was liable for the blade failures. [holding]
- The accident report found the pilots not at fault. [holding]
'''
```

Your Task:

QUESTION: [QUESTION_PLACEHOLDER]

ATOMIC CLAIMS:

Instruction 3(c)–(e) and instruction 4 are the extension. LOREFACT’s Table 12 asks for 10 correct and 10 incorrect claims and nothing else; its instruction that claims be “independently verifiable” actively *precludes* the equivalent pairs an EQUIVALENCE edge needs. The tagged pairs are the raw material for the three couplings with no prior gold, and the [holding] claims are what make a *resolved* Concession constructible. Their required vocabulary is not stylistic: it matches the cue list in the miner’s `looks_like_holding` heuristic (held, ruled, concluded, tribunal, determined, ...), so a planned resolution is one the shipped resolver can actually find. Instruction 5 is load-bearing in the other direction: claims must arrive *connective-free* so that every relation in the final response traces to P3’s plan rather than to an accident of claim phrasing.

A.3 P3 — relation plan generation

P3 — relation plan generation extnormal(the architectural departure, Section 6.1)

Inputs: [QUESTION_PLACEHOLDER], [CLAIMS_PLACEHOLDER] (P2s tagged list).

Output: One JSON object: atoms (with pos), relations (sense, strength band, validity, error kind, resolver), non_relations.

Gate: Schema-valid; every pair within 4 positions or sharing a named entity (instr. 4); ≥ 1 each of Alternative, Disjunction, Restatement, Concession and Precedence/Succession (instr. 5); 55/45 valid/invalid (instr. 8).

This replaces LOREFACT’s logical-statement stage (their Table 13). Their stage emits prose and requires that “each logical statement must contain a logical error”, which is what produces their 78% skew. Ours emits a graph and controls the valid/invalid ratio explicitly.

Instructions:

1. You are given a QUESTION and a tagged list of ATOMIC CLAIMS. Produce a RELATION PLAN: a machine-readable graph of the logical relations a response should draw between those claims.
2. Select 14-18 claims to use, and assign each a POSITION index 1..N giving the order in which the response will assert it. Every selected claim keeps its text EXACTLY as given. Do NOT introduce information not in the claims, and do NOT alter the claims.
3. Plan 8-12 RELATIONS. Each relation connects exactly two selected claims and has a SENSE, one of: Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast, Concession, Alternative, Disjunction, Precedence, Succession.
 - Cause-Effect: source causes target. Effect-Cause: source is the effect, target its cause.
 - Evidence: source is evidence for target. Condition: source is a condition for target.
 - Restatement: source and target assert the same thing.
 - Instantiation: source is a general claim, target a specific instance.
 - Contrast: opposition that is NOT exhaustive - both could be false.
 - Concession: a tension the text itself resolves.
 - Alternative: EXHAUSTIVE competing options - EXACTLY ONE holds. Not both, and not neither. Use your [alt-pair-*] claims here.
 - Disjunction: AT LEAST ONE holds; both may hold. Use your [disj-pair-*] claims here.
 - Precedence / Succession: ordered in time with NO truth dependence - neither claim being true or false says anything about the other.
4. Every relation’s two claims must be within 4 POSITIONS of each other, OR must share a named entity appearing in both claim texts. This keeps the graph recoverable from the response by a local reader.
5. The plan must include AT LEAST ONE relation of each of these senses: Alternative, Disjunction, Restatement, Concession, and one of Precedence/Succession.
6. Every Concession must be marked resolved or unresolved. A resolved Concession must name a RESOLVER: a [holding] claim, also selected and positioned AFTER both endpoints.
7. Give each relation an intended STRENGTH BAND: strong (0.85-1.00), moderate (0.60-0.84), or weak (0.35-0.59) - how strongly the source determines the target under that sense. Restatement relations are always strong.
8. Give each relation an intended VALIDITY: valid if the relation genuinely holds between the claims as stated; invalid if it does not. Plan 55% valid

- and 45% invalid. For each invalid relation give an `ERROR_KIND`, one of: `wrong_sense`, `wrong_direction`, `false_endpoint`, `spurious`.
- Also list 4-6 `NON-RELATIONS`: pairs of selected claims within 4 positions of each other between which the response must draw NO logical dependence. These are the distractors a relation miner must correctly label `None`.
 - Output a single JSON object in a code block, exactly in this schema, and nothing else:

```

'''json
{
  "atoms": [{"pos": 1, "text": "...", "factual": true}],
  "relations": [
    {"source_pos": 1, "target_pos": 3, "sense": "Cause-Effect",
     "strength_band": "strong", "validity": "valid",
     "error_kind": null, "resolved": null, "resolver_pos": null}
  ],
  "non_relations": [{"source_pos": 2, "target_pos": 5}]
}
'''

```

Your Task:

QUESTION: [QUESTION_PLACEHOLDER]

ATOMIC CLAIMS: [CLAIMS_PLACEHOLDER]

RELATION PLAN:

Instruction 4 is the recoverability constraint and exists because of R2 (Section 9): the default candidate policy is `gated` with `window=4`, so a planned edge whose endpoints are farther apart and which fails both the similarity gate and the discourse-adjacency check is *unmineable by construction* and cannot be a fair miner error. Instruction 5 guarantees per-item coverage of the no-prior-gold facets. Instruction 9 is what makes the over-connection rate measurable at all.

A.4 P4 — response generation

P4 — response generation

Inputs: [QUESTION_PLACEHOLDER], [PLAN_PLACEHOLDER] (P3s JSON).

Output: The response prose in a code block, ≥ 500 words.

Gate: v1 (relations recoverable), v3 (no leakage, no hedging), v4 (100% of atoms asserted).

Instructions:

- You are given a `QUESTION` and a `RELATION PLAN` in JSON. Write a `RESPONSE` that answers the question and realizes the plan exactly.
- Assert every planned atom, in the plan's `POSITION` order, preserving each atom's factual content. You may adjust surface wording for fluency but must not change what is asserted, and must not merge two atoms into one clause.
- Realize every planned `RELATION` so a careful reader could recover it from the text alone, using language appropriate to its sense:
 - Cause-Effect / Effect-Cause: "caused", "led to", "resulted from", "because"
 - Evidence: "indicates", "confirms", "as shown by"
 - Condition: "if", "provided that", "only when"
 - Restatement: "that is", "in other words", "equivalently"
 - Instantiation: "for example", "specifically", "in one case"
 - Contrast: "by contrast", "whereas", "on the other hand"
 - Concession: "although", "despite", "even though"

- Alternative: make the exhaustiveness EXPLICIT - "either ... or", "one of these accounts must be wrong", "the two cannot both be true, and one of them must hold"
 - Disjunction: make the at-least-one reading EXPLICIT - "at least one of", "one or both"
 - Precedence / Succession: "before", "after", "subsequently" - and make clear the ordering carries NO causal or inferential force. Do NOT write "and therefore" or "which led to" around these.
4. For each planned NON-RELATION, assert both atoms but draw no dependence between them. Separate them clearly - use "Separately,", "In an unrelated matter,", or place them in different paragraphs. Do not let a connective bridge them.
 5. For a resolved Concession, assert the resolver atom after both endpoints and present it as settling the tension.
 6. Use precise and formal language, avoiding vague generalizations and rhetorical fillers. Exhibit clear logical organization. No fewer than 500 words.
 7. Do NOT mention the relation plan, the senses, the strength bands, the validity labels, or the fact that some content is incorrect. Do NOT hedge with "assume", "might", "allegedly", "supposedly", or "reportedly" around planned-invalid relations - assert them in the same register as the valid ones. Do NOT add headings, bullet lists, or annotations.
 8. Wrap the RESPONSE in a code block.

Your Task:

QUESTION: [QUESTION_PLACEHOLDER]

RELATION PLAN: [PLAN_PLACEHOLDER]

RESPONSE:

Instruction 3's per-sense vocabulary is what makes the gold graph recoverable; the explicit exhaustiveness requirement for Alternative and Disjunction is what makes the CONTRADICTION/EXCLUSIVE/CO-NECESSITY distinction present *in the text* rather than only in the annotation. Instruction 7 is the anti-leakage clause, verified by V3. Its second half matters more than it looks: if the generator hedges around planned-invalid relations, a miner can detect invalidity from register alone and the benchmark measures hedge-detection instead of reasoning — LOREFACT guards the same risk in their Table 13 instruction 5.

A.5 P5 — parameterized perturbation

P5 — parameterized perturbation

Inputs: [RESPONSE_PLACEHOLDER], [PLAN_PLACEHOLDER], and [OPERATOR_PLACEHOLDER] — filled with *one* of the ten call expressions of instruction 3, e.g. `wrong_sense(r003, "Contrast")` or `shuffle_order(adjacent)`. An edge argument is a `relations[].id` from the plan.

Output: Two code blocks: the perturbed response, then the JSON diff.

Gate: Exactly one operator applied; length within 15% (except the three operators that add or remove a sentence); all other relations preserved verbatim.

One prompt covering all ten operators, so ladders are reproducible and the diff is machine readable.

Instructions:

1. You are given a RESPONSE, its RELATION PLAN, and one PERTURBATION to apply.
2. Apply ONLY the named perturbation. Change the minimum number of words

- needed. Every other relation, atom, and sentence must be preserved verbatim.
3. The perturbations are defined as follows.
 - `wrong_sense(edge, new_sense)`: rewrite the connective realizing that edge so the text draws `new_sense` instead. Keep both atoms' content unchanged.
 - `direction_reversal(edge)`: rewrite so the relation runs from the target to the source. Keep both atoms' content unchanged.
 - `exhaustiveness_flip(edge)`: if the edge is `Alternative`, rewrite so the two claims merely oppose without being exhaustive (both could be false). If it is `Contrast`, rewrite so exactly one must hold.
 - `remove_resolution(edge)`: delete the resolver sentence and any reference to it, leaving the conceded tension unresolved. Change nothing else.
 - `inject_contradiction(atom)`: add ONE new sentence asserting a claim that directly contradicts the named atom. Place it at least two sentences later.
 - `break_chain(edge)`: remove or negate the link realizing that edge so the causal chain no longer connects, keeping both endpoint atoms present.
 - `shuffle_order(level)`: reorder the response's sentences without changing any wording. `level=full` permutes all sentences; `level=block` permutes paragraph blocks; `level=adjacent` swaps one adjacent pair.
 - `spurious_relation(atom_a, atom_b)`: add a connective asserting a causal or contradictory link between two atoms the original response kept separate. Add no new factual content.
 - `drop_relation(edge)`: remove the connective realizing that edge so the two atoms are merely adjacent assertions. Keep both atoms.
 - `ordering_only(edge)`: rewrite a `Precedence` edge as `Succession`, or the reverse, preserving the actual chronology and adding NO causal or inferential language. The truth of neither atom may become dependent on the other.
 4. Do not change the response's length by more than 15 percent, except for `inject_contradiction` (which adds one sentence) and `remove_resolution` / `break_chain` (which remove one).
 5. Output the perturbed RESPONSE in a code block, then a second code block containing a JSON diff:


```
{
  "operator": "...",
  "target": ...,
  "sentences_changed": [...],
  "relations_added": [...],
  "relations_removed": [...],
  "relations_relabeled": [...]}

```

Your Task:

RESPONSE: [RESPONSE_PLACEHOLDER]

RELATION PLAN: [PLAN_PLACEHOLDER]

PERTURBATION: [OPERATOR_PLACEHOLDER]

The `ordering_only` operator's final sentence is the whole content of the invariance test of Section 4.3: if the rewrite introduces any inferential language, the edit is no longer factor-neutral and the test is void.

A.6 V1 — round-trip relation recovery

V1 — round-trip relation recovery extnormal(the load-bearing gate)

Inputs: [RESPONSE_PLACEHOLDER], [ATOMS_PLACEHOLDER]. The plan is deliberately withheld.

Output: A JSON list of recovered relations: source, target, sense, coupling, strength band, resolution.

Gate: Admission: a majority of the committee must recover $\geq 80\%$ of planned relations with the correct coupling and $\geq 70\%$ with the correct sense. Unrecovered edges are dropped or the item regenerated — never kept as gold.

The acceptance gate. An independent model, shown the response and its atoms but *not* the plan, re-derives the relation graph; an item enters the benchmark only if its planned graph is actually recoverable.

Let's define a function named `recover_relations(response: str, atoms: list[str])`. Given a RESPONSE and the list of ATOMS drawn from it, the function returns the list of logical relations the response ITSELF draws between those atoms.

For each ORDERED pair (i, j) that the response actually relates, emit one object:

```
{
  "source": i, "target": j,
  "sense": one of ["Cause-Effect", "Effect-Cause", "Evidence", "Condition",
    "Restatement", "Instantiation", "Contrast", "Concession", "Alternative",
    "Disjunction", "Precedence", "Succession"],
  "coupling": one of ["entailment", "contradiction", "equivalence", "exclusive",
    "co_necessity", "none"],
  "strength_band": one of ["strong", "moderate", "weak"],
  "resolved": true or false or null}
```

Rules:

- Emit a relation ONLY if the response itself draws that connection, as written or as a clear step in the author's argument. Do not emit a relation that is merely plausible in general.
- Use "Alternative" only when the two claims cannot both be true AND cannot both be false. Use "Contrast" when they merely cannot both be true.
- Use "Disjunction" only when the response rules out both claims being false.
- Use "Precedence" or "Succession" only when the ordering carries no truth dependence; set their coupling to "none".
- Pairs the response leaves unrelated must be omitted entirely.

The function returns a JSON list. Note that your response will be passed to the python interpreter, SO NO OTHER WORDS!

```
recover_relations(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])
```

The gate. An item is admitted only if a majority of the V1 committee recovers $\geq 80\%$ of its planned relations with the correct *coupling* and $\geq 70\%$ with the correct *sense*. Unrecovered edges are dropped from gold or the item is regenerated — **never** retained as unrecoverable gold, which would penalize a correct miner for failing to read something that is not in the text.

A.7 V2 — exhaustiveness adjudication

V2 — exhaustiveness adjudication

Inputs: [CLAIM_A_PLACEHOLDER], [CLAIM_B_PLACEHOLDER], [RESPONSE_PLACEHOLDER]. Run on v1s conflict edges.

Output: One letter: A exhaustive, B non-exhaustive, C disjunction, D neither.

Gate: **Unanimity** of the committee is required to assign an EXCLUSIVE gold label; a split vote ships the edge low-agreement (Section 9, R1).

The Contrast/Alternative boundary is the design's hardest label (Section 3.4). This prompt makes it a *procedure* rather than a matter of annotator taste.

Let's define a function named `adjudicate_exhaustiveness(claim_a: str, claim_b: str, response: str)`.

Given two claims and the response that asserts both, the function decides the logical relationship between their truth values, returning exactly one of:

- "A" - EXHAUSTIVE EXCLUSION: exactly one of A and B holds. They cannot both be true, and they cannot both be false.
- "B" - NON-EXHAUSTIVE OPPOSITION: they cannot both be true, but both could be false (some third possibility exists).
- "C" - DISJUNCTION: at least one holds, and both may hold together.
- "D" - NEITHER: the truth values are not coupled in any of these ways.

To decide between A and B, ask: is there a coherent third state of the world in which A and B are BOTH false? If yes, answer B. If no such third state exists, answer A.

To decide between A and C, ask: could A and B both be true at once? If yes, answer C.

Answer with the single letter only. Note that your response will be passed to the python interpreter, SO NO OTHER WORDS!

```
adjudicate_exhaustiveness(claim_a="[CLAIM_A_PLACEHOLDER]",
                           claim_b="[CLAIM_B_PLACEHOLDER]",
                           response="[RESPONSE_PLACEHOLDER]")
```

The “coherent third state” test is the operative decision procedure, and it maps directly onto the factor tables: a third state where both are false is exactly the (0,0) world that EXCLUSIVE penalizes and CONTRADICTION permits.

A.8 V3 — naturalness and leakage audit

V3 — naturalness and leakage audit

Inputs: [RESPONSE_PLACEHOLDER].

Output: JSON: fluency/formality/organization (1-5), plus leakage, hedging, artifacts span lists.

Gate: All three scores ≥ 4 , and leakage and hedging both empty.

Let's define a function named `check_response(response: str)`.

The function audits a generated response and returns a JSON object:

```
{"fluency": 1-5, "formality": 1-5, "organization": 1-5,
 "leakage": [list of quoted spans], "hedging": [list of quoted spans],
 "artifacts": [list of quoted spans]}
```

- fluency, formality, organization: 5 = indistinguishable from expert human prose; 1 = badly degraded.
- leakage: any span revealing the response was generated from an annotation plan - naming a relation type or sense, referring to "claims", "atoms", "statements", or a plan; any label, tag, index, or bracketed marker.
- hedging: any span using "assume", "might", "possibly", "allegedly", "supposedly", "reportedly", or "it is claimed", which would signal to a reader which content is unreliable.
- artifacts: enumerated list structure, headings, repeated template phrasing, or a sentence that reads as a stitched-together claim rather than prose.

Return the JSON object only. Note that your response will be passed to the

python interpreter, SO NO OTHER WORDS!

```
check_response(response="[RESPONSE_PLACEHOLDER]")
```

Gate: all three scores ≥ 4 , and leakage and hedging both empty.

A.9 V4 — atom coverage and realized positions

V4 — atom coverage and realized positions

Inputs: [RESPONSE_PLACEHOLDER], [ATOMS_PLACEHOLDER].

Output: Per atom: status (asserted/altere/missing/merged), span, and sentence position.

Gate: 100% asserted. The position output re-verifies P3s window constraint against the realized text (Section 9, R2).

Let's define a function named `check_atom_coverage(response: str, atoms: list[str])`.

For each atom, the function decides whether the response asserts it, returning a JSON list of objects:

- ```
{
 "index": i,
 "status": one of ["asserted", "altere", "missing", "merged"],
 "span": the quoted span of the response that asserts it, or null,
 "position": the 1-based index of that span among the response's sentences}

```
- "asserted": the response asserts the atom's full content, in any wording.
  - "altere": the response asserts something related but changes the atom's content (a different number, date, entity, or polarity).
  - "missing": the response does not assert the atom.
  - "merged": the atom's content is fused into a clause that also asserts another atom, so the two cannot be separated.

Return the JSON list only. Note that your response will be passed to the python interpreter, SO NO OTHER WORDS!

```
check_atom_coverage(response="[RESPONSE_PLACEHOLDER]", atoms=[ATOMS_PLACEHOLDER])
```

Gate: 100% asserted. The position field has a second job — P3 constrains *planned* positions, but P4's realized sentence order may drift, so the window constraint must be re-verified against realized positions post hoc (Section 9, R2).